ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

Enthusiast Course

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL 2022

Test Type :**PRACTICE TEST** Test Pattern :**NEET (UG)**

TEST DATE :

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	2	1	1	2	2	3	1	2	3	1	1	1	2	2	2	4	4	2	3	1	3	2	2	1	4	3	4	3	3

Q.	31	32	33	34	35	36	37	38	39	40	41	42	43
A.	4	3	4	3	3	3	2	3	4	2	3	4	3

HINT – SHEET

1. Ans (4)

$E_n \propto \frac{1}{n^2}$ and $r_n \propto n^2$

So, $E_n r_n$ is independent of n

Hence, $E_1 r_1 = (13.6 \text{ eV})(0.53 \text{ Å})$

7.2 eV Å = constant

2. Ans (2)

From einstein's photoelectric effect concept the energy of these photons, for light of frequency v is E = hυ

Where h is plank's constant.

Also, frequency $= \frac{\text{velocity}}{\text{wavelength}} = \frac{c}{\lambda} = \frac{hc}{\lambda}$

Energy of n photons is E = nhc/λ

Given $E=10^{-7}$ J , $\lambda = 5000 \text{ Å} = 5000 \times 10^{-10}$ m

$$n = \frac{E\lambda}{hc} = 2.5 \times 10^{11}$$

3. Ans (1)

For hydrogen like atom,

$$\frac{1}{\lambda'} = Rz^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \quad .....(i)$$

For hydrogen atom,

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \quad .....(ii)$$

Dividing eqn (ii) by (i) we get

$$\frac{\lambda}{\lambda'} = \frac{\left(Rz^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)\right)}{\left(R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)\right)}$$

$$\frac{\lambda}{\lambda'} = z^2$$

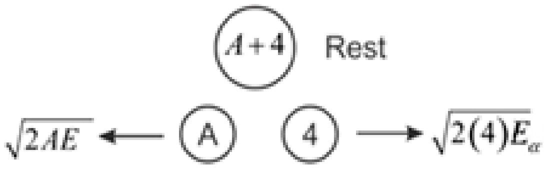
According to problem,

$$\lambda' = \frac{\lambda}{4}$$

$\Rightarrow z^2 = 4$

z = 2

4. Ans (1)



By momentum conservation

$$\sqrt{2AE} = \sqrt{2\,(4)\,E\alpha}$$

$$E\alpha = \frac{A}{4}E$$

So energy released

$$E_R = E + \frac{A}{4}E = \frac{E}{4}(A+4)$$

5. Ans (2)

$$K.E. = \frac{KZe^2}{2r}$$

$$P.E. = \frac{-KZe^2}{r}$$

$$T.E. = P.E. + K.E. = \frac{-KZe^2}{2R}$$

Therefore, TE = –KE = PE/2 = – 3.4 eV

So, KE = 3.4 eV

Let p = momentum and m = mass of the electron

$E = p^2/2m$

De Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

On substituting the value, we get

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 3.4 \times 1.6 \times 10^{-19}}}$$

$$= 6.6 \times 10^{-10} m$$

6. Ans (2)

Given, $E = 3eV = 3 \times 1.6 \times 10^{-19}$ J

We know that $\lambda = \dfrac{h}{\sqrt{2mE}}$

$$= \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 1.7 \times 10^{-27} \times 3 \times 1.6 \times 10^{-19}}}$$

$$= 1.65 \times 10^{-11}\;m$$

7. Ans (3)

$$wE_n = -13.6\,\frac{Z^2}{n^2}$$

$$\Delta E = E_4 - E_2$$

$$= -13.6\,z^2\left[\frac{1}{16} - \frac{1}{4}\right]$$

= 2.55 eV

8. Ans (1)

In beta decay energy mission is continuous and energy is shared between beta particles and daughter nuclei. thus, certain beta particle has zero energy and certain have maximum energy therefore option (1) is most appropriate.

9. Ans (2)

The number of photoelectrons decided the photocurrent. Assuming that the number of electrons emitted depends on the number of photons incident, the number of photoelectrons depend on the intensity of light.

10. **Ans (3)**

$^2_1He + \, ^2_1He \, ^4_2He + \text{energy}$

Energy released

$= \text{B.E. of} \, ^4_2He - 2(\text{B. E. of} \, ^2_1He)$

= 28 – 2(2.2)

= 28 − 4.4

= 23.6 MeV

11. **Ans (1)**

Modified equation of de-Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$\lambda = \frac{12.27 \, \overset{o}{A}}{\sqrt{V}}$$

$$= \frac{12.27}{\sqrt{64}} = 1.534\overset{o}{A}$$

12. **Ans (1)**

$$\frac{1}{2}mv^2 = hf - \phi_0 = hf - hf_0$$

The kinetic energy of the emitted photoelectrons is distributed from zero to the maximum value. Minimum kinetic energy of emitted photoelectron is zero.

13. **Ans (1)**

N_0 is the initial amount of substance and N is the amount left after decay

Thus,

$$N = N_0 \left(\frac{1}{2}\right)^n$$

N = no. of half-lives

$$= \frac{t}{t_{1/2}} = \frac{15}{5} = 3$$

Therefore,

$$N = N_0 \left(\frac{1}{2}\right)^3 = \frac{N_0}{8}$$

14. **Ans (2)**

Use the relationship between energy of an orbit and quantum number n of that orbit.

$$\frac{r_4 - r_3}{r_9 - r_8} = \frac{r_1 \times 4^2 - r_1 \times 3^2}{r_1 \times 9^2 - r_1 \times 8^2}$$

$$= \frac{16 - 9}{81 - 64} = 0.41$$

15. **Ans (2)**

The number of photoelectrons ejected is directly proportional to the intensity of incident light. Maximum kinetic energy is independent of intensity of incident light but depends upon the frequency of light. Hence, option 2N and T is correct.

16. **Ans (2)**

Number of emitted γ-rays per second is proportional to no of uranium particles left undecayed

Fraction of particles undecayed =

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^n$$

Where n is the number of half-lives

$$\frac{5}{20} = \left(\frac{1}{2}\right)^n$$

n = 2

Total time = $2 \times 5 \times 10^9$ years $= 10^{10}$ years

17. **Ans (4)**

$$\frac{\lambda_1}{\lambda_2} = \frac{\left(\frac{1}{9} - \frac{1}{16}\right)}{\left(\frac{1}{1} - \frac{1}{4}\right)} = \frac{7}{108}$$

18. **Ans (4)**

Radius (R) of nucleus is related with mass number (A) as $R \propto A^{1/3}$

Now, $R^3 \propto A$

$\Longrightarrow$ Volume $\propto$ A

19. Ans (2)

Initial momentum = E/c

Final momentum = –E/c

So,

Change of momentum = E/c – (–E/c) = 2E/c

Momentum transferred to surface = 2E/c

20. Ans (3)

Emission of one alpha and two beta particles won't change the atomic number and hence same element will be the final product.

21. Ans (1)

$h\nu_0 = 6.2$ eV, $e\nu_0 = 5$ eV

From Einstein's photoelectric equation

$$h\nu = h\nu_0 + eV_0 = 6.2 + 5 = 11.2 \text{ eV}$$

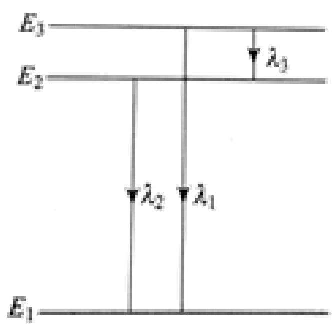
$$\frac{hc}{\lambda} = 11.2$$

$$\lambda = \frac{hc}{11.2} = 1108.9 \text{ Å}$$

it belongs to ultra violet region

22. Ans (3)

Let the three energy levels be E_1, E_2 and E_3. The wavelength λ_1, λ_2 and λ_3 of the spectra lines corresponding to the three energy transitions.



$$E = hf = \frac{hc}{\lambda};$$

Given that $\lambda_1 < \lambda_2 < \lambda_3$

Thus, for the three wavelengths, we have

$$E_3 - E_2 = \frac{hc}{\lambda_3} \quad .....(1)$$

$$E_2 - E_1 = \frac{hc}{\lambda_2} \quad .....(2)$$

$$E_3 - E_1 = \frac{hc}{\lambda_1} \quad .....(3)$$

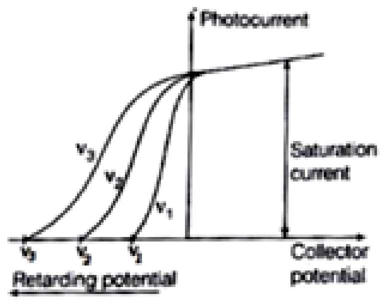
$$E_3 - E_1 = \left(E_3 - E_2\right) + \left(E_2 - E_1\right)$$

$$\frac{hc}{\lambda_1} = \frac{hc}{\lambda_3} + \frac{hc}{\lambda_2}$$

$$\frac{1}{\lambda_1} = \frac{1}{\lambda_3} + \frac{1}{\lambda_2}$$

27. Ans (3)

The smaller the incident frequency, the smaller the stopping voltage required.



$\nu_1 < \nu_2 < \nu_3$

28. Ans (4)

The magnitude of momentum and de-Broglie is related as $\lambda = \frac{h}{p}$.

Therefore, if linear momenta are same then wavelengths are equal but if wavelengths are same then only the magnitudes of momenta are same while direction might be different

29. Ans (3)

1 amu = 1.66×10^{-27} kg

Equivalent energy can be calculated by using mass-energy equivalence equation

$E = mc^2$

$E = 1.66 \times 10^{-27} \times (3 \times 10^8)^2 J$

$1 MeV = 10^6 \times 1.6 \times 10^{-19} J$

Using about equations we can get,

E = 931 MeV

30. Ans (3)

Cadmium rods are used in the form of control rod. In a nuclear reactor, it can absorb the neutrons and are used to control the nuclear Chain Reaction. Even Boron rods are used for this purpose.

31. Ans (4)

$\lambda_p = \lambda_\alpha$

$$\frac{h}{\sqrt{2m_p Q_p V_p}} = \frac{h}{\sqrt{2m_\alpha Q_\alpha V_\alpha}}$$

$\therefore m_p Q_p V_p = m_\alpha Q_\alpha V_\alpha$

$$\therefore V_\alpha = \left(\frac{m_p}{m_\alpha}\right)\left(\frac{Q_p}{Q_\alpha}\right)$$

$$V_\alpha = \left(\frac{1}{4}\right)\left(\frac{1}{2}\right) V_p = \frac{V}{8}$$

32. Ans (3)

The energy of an electron in hydrogen atom is

$$E = -\frac{13.6}{n^2} eV$$

So, the energy required to remove an electron from n = 2 state is

$$|E| = \frac{13.6}{2^2} eV$$

$$= \frac{13.6}{4} = 3.4 eV$$

33. Ans (4)

The de-Broglie wavelength of a particle is $\lambda = \frac{h}{p}$, where *h* is Plank's constant and *p* is a momentum of the particle. So, the curve will be a rectangular hyperbola.

34. Ans (3)

Let number of α particles decayed be x and number of β particles decayed be y. Then equation for the decay is given by

$${}_{92}U^{235} \rightarrow x\alpha_2^4 + y\beta_{-1}^0 + Pb_{82}^{203}$$

Equating the mass number on both sides

235 =4x 203 . .....(i)

Equating atomic number on both sides

92 = 2x – y + 82 .....(ii)

Solving Equ.(i) and(ii). we get

x = 8, y = 6

∴ 8 α particles and 6 β particles are emitted in disintegration.

35. **Ans (3)**

$K_1 = \frac{hc}{\lambda_1} - W$ .... (i)

and $K_2 = \frac{hc}{\lambda_2} - W$ .... (ii)

Substituting $\lambda_1 = 2\lambda_2$ in Eq. (i), we get

$K_1 = \frac{hc}{2\lambda_2} - W$

$= \frac{1}{2}\left(\frac{hc}{\lambda_2}\right) - W = \frac{1}{2}(K_2 + W) - W$

$\therefore K_1 = \frac{K_2}{2} - \frac{W}{2}$

or $K_1 < \frac{K_2}{2}$

36. **Ans (3)**

When a nucleus is formed, then the mass of nucleus is slightly less than the sum of Z protons and N neutrons. That difference in energy also known as binding energy needs to be supplied to nucleus to break it into respective number of protons and neutrons.

$M < Zm_p + Nm_n$

37. **Ans (2)**

Work function is given by

$\phi = \frac{hc}{\lambda}$ or $\phi \propto \frac{1}{\lambda}$

$\therefore \frac{\phi_1}{\phi_2} = \frac{\lambda_1}{\lambda_2} = \frac{600}{300} = 2$

38. **Ans (3)**

Effective mean life for parallel disintegration is given by

$\tau = \frac{\tau_1 \tau_2}{\tau_1 + \tau_2}$

$= \frac{1620 \times 660}{2280} = 469$ year

Let t be the time in which $\left(\frac{1}{4}\right)^{th}$ of the material remains intact

$\frac{N}{4} = \frac{N}{2^{\frac{t}{\tau}}}$

$t = 2\tau \ln 2$

$= 2 \times 469 \times 0.693 = 650$ years

39. **Ans (4)**

If r_0 is radius of first orbit in hydrogen atom then radius of first orbit in hydrogen like atoms is given by

$r = \frac{r_0}{z}$

For hydrogen, Z = 1

For deuterium, Z = 1

For singly ionized helium He^+, Z = 2

For doubly ionized lithium Li^{2+}, Z = 3

Radius of the first orbit, r is minimum for doubly ionized lithium.

40. **Ans (2)**

After one hour $\frac{A_0}{\sqrt{3}A_0} = \frac{1}{\sqrt{3}} = \left(\frac{1}{2}\right)^{\frac{1}{T}}$

After next 3 hours means four hours from initial

$\frac{A}{A_0} = \left(\frac{1}{2}\right)^{\frac{4}{T}} = \left[\left(\frac{1}{2}\right)^{\frac{1}{T}}\right]^4 = \left[\frac{1}{\sqrt{3}}\right]^4 = \frac{1}{9}$

41. **Ans (3)**

From conservation of mechanical energy

$\Delta KE = q(V_i - V_f)$

$\frac{K_\alpha}{K_P} = \frac{2eV}{eV} = 2$

42. **Ans (4)**

We know that

$E = \phi + KE$

$\Rightarrow h\nu = W + eV_0$

$\Rightarrow eV_0 = h\nu - W$

$V_0 = \frac{h}{e}\nu + \left(-\frac{W}{e}\right)$

Compare this with equation of line y = mx + c

$m = \text{slope of graph} = \frac{h}{e}$

43. Ans (3)

Radioactive decay equation is

$N = N_0 e^{-\lambda t}$

$$\ln\left(\frac{N}{N_0}\right) = -\lambda t = -\frac{\ln 2 \times t}{T_{1/2}}$$

After decay of 99% of the initial sample only 1% will be left.

$$\frac{N}{N_0} = \frac{1}{100}$$

$$\Rightarrow -\ln 100 = -\frac{\ln 2 \times t}{T_{1/2}}$$

$$\therefore t = \frac{\ln 100}{\ln 2} \times T_{1/2}$$

$$= \frac{4.606}{0.693} \times 110$$

$$= 731\text{min} = 12.2\text{h}$$